Mohammad Amin Hasanpour

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Ph.D. researcher at DTU working on **Embedded Artificial Intelligence (eAI)** with a focus on **TinyML frameworks, benchmarking, and quantization techniques**. Published in peer-reviewed venues and contributed to open-source projects **bridging deep learning and embedded systems**. Experienced in teaching, collaborative research, and leading technical projects. Motivated to contribute to both academia and industry by making **AI more efficient, scalable, and deployable on edge devices**.

Education

Ph.D. **Technical University of Denmark (DTU)** — Lyngby, Denmark
Applied Mathematics and Computer Science — Jan 2023 - (Jan 2026)
Thesis: Embedded Artificial Intelligence (eAI)
Supervisor: Dr. Xenofon Fafoutis
*In close collaboration with **Grundfos** and **VELUX***

M.Sc. **Sharif University of Technology** — Tehran, Iran
Electrical Engineering — Sep 2019 - Sep 2021
Thesis: Improvement of Deep Learning-Based Coding Algorithms
Supervisor: Dr. Alireza Farhadi
Cumulative GPA: 4.0/4.0, Thesis grade: Excellent

B.Sc. **K. N. Toosi University of Technology** — Tehran, Iran
Electrical Engineering — Sep 2015 - Sep 2019
Thesis: Design and Fabrication of an IoT Platform
Supervisor: Dr. Yousef Darmani
Cumulative GPA: 3.67/4.0, Thesis grade: Excellent

Domain Expertise

TinyML, **Deep Learning**, Machine Learning, Embedded Systems

Skills

Proficient in	**TensorFlow, Keras, Python, C++**
Experienced in	**Software**: AI-assisted development, PyTorch, ONNX, Git, Conda, Docker, DVC, Linux, Mkdocs
	Hardware: VHDL, Verilog, MCU programming (AVR, STM32, Arduino, ESP8266), PCB design

Publications

A Survey of Quantization Techniques in Embedded AI Toolchains — Dec 2025
M.Amin Hasanpour, Xenofon Fafoutis, Manuel Roveri
Accepted at IEEE AIoT Conference 2025

EdgeMark: An Automation and Benchmarking System for Embedded Artificial Intelligence Tools — Oct 2025
M.Amin Hasanpour, Mikkel Kirkegaard, Xenofon Fafoutis
10.1016/j.sysarc.2025.103488

A Holistic Review of the TinyML Stack for Predictive Maintenance — Dec 2024
Emil Njor, ***M.Amin Hasanpour***, Jan Madsen, Xenofon Fafoutis
10.1109/ACCESS.2024.3512860

Pump Cavitation Detection with Machine Learning: A Comparative Study of SVM and Deep Learning Jul 2024
M.Amin Hasanpour, Rasmus Engholm, Xenofon Fafoutis
10.1109/AIoT63253.2024.00050

Selected Projects

EdgeMark GitHub
- Developed an automation system to generate, convert, compile, and test models on embedded devices.
- Supported multiple model types, four eAI tools (e.g., TFLM), and two microcontrollers (Cortex-M4, RX65N).
- Conducted extensive benchmarking to compare tool performance under varying conditions.

Cavitation Detection in Centrifugal Pumps GitHub
- Implemented and benchmarked SVM vs. deep learning approaches on real industrial data.
- Optimized models for deployment on embedded hardware.

Improvement of Deep Learning-Based Coding Algorithms
- Reduced network parameter count by 40% without compromising performance.
- Designed improved architectures for deep learning-based coding.

Cassava Leaf Disease Classification - Kaggle Competition
- Applied architectures including ResNet, Inception-ResNet-v2, VGG16, and EfficientNet.
- Applied CycleGAN, CGAN, data pipelines, test-time augmentation, and k-fold cross-validation.
- Achieved the best performance in the class project competition.

Implementation of MS-VAN for NWPU VHR-10 Object Detection
- Built MS-VAN model integrating Faster R-CNN, skip-connected autoencoders, and attention modules.
- Evaluated performance on high-resolution object detection tasks.

Generative Adversarial Networks (GANs) Adventure
- Explored multiple GAN variants including DCGAN, CycleGAN, and LSTM-based GAN architectures.
- Designed and trained a music-composer GAN using LSTM layers.
- Achieved stable DCGAN performance generating MNIST-like images.

Cartpole Racing with Reinforcement Learning
- Built a reinforcement learning agent using policy gradient methods.
- Trained the agent to balance a pole and race in a custom environment.

Implementation of Deep JSCC Image Transmission
- Developed an autoencoder-based joint source-channel coding (JSCC) method for image transmission.
- Outperformed traditional JPEG/JPEG2000 + channel coding under specific conditions.

Generating Images Using Autoencoders
- Compared performance of simple autoencoders, VAEs, and disentangled VAEs.
- Conducted experiments on the MNIST dataset to analyze the performance of different models.

Design and Fabrication of an IoT Platform Webpage
- Designed and fabricated gateway and sensor boards with PCB design and circuit integration.
- Configured a backend server to collect IoT data.
- Developed an Android application to visualize user's data.

BuyNow - Smart Store (Internship at HardTech) Website
- Implemented local login option and improved system security.
- Migrated communication protocol from MQTT to HTTP.
- Developed auto-update and debugging systems and enhanced the user interface.

Teaching Experience

Master Thesis Supervision: Neural Architecture Generation for Edge AI — Jan 2025 - Jun 2025
Student: William Diedrichsen Marstrand - DTU Compute

Project Course Supervision: Neural Network Architecture for Resource Constrained Hardware — Oct 2024 - Dec 2024
Student: William Diedrichsen Marstrand - DTU Compute

Project Course Supervision: TinyML - Efficient Machine Learning for Embedded AI Systems — Jun 2024
Student: William Diedrichsen Marstrand - DTU Compute

Project Course Supervision: Embedded Programming for Benchmarking Embedded AI Models — Feb 2024 - May 2024
Student: Simon Kristoffer Janum - DTU Compute

Teaching Assistant: Deep Learning — Sep 2023 - Nov 2023
Instructor: Dr. Jes Frellsen - DTU Compute

Teaching Assistant: Deep Learning — Sep 2021 - Jan 2022
Instructor: Dr. Emad Fatemizadeh - Sharif University of Technology

Teaching Assistant: MLOps — Jan 2025
Instructor: Dr. Nicki Skafte Detlefsen - DTU Compute

Teaching Assistant: Operating Systems — Sep-Nov 2023, 2024
Instructor: Dr. Xenofon Fafoutis - DTU Compute

Teaching Assistant: Parallel Programming and Architectures — Jan 2021 - Jul 2021
Instructor: Dr. Matin Hashemi - Sharif University of Technology

Teaching Assistant: Medical Image Analysis and Processing — Jan 2021 - Jul 2021
Instructor: Dr. Emad Fatemizadeh - Sharif University of Technology

Achievements and Awards

- Received dual research grants to support a research stay at Politecnico di Milano (2025, Milan)
  Thomas B. Thriges, IDA (Danish Society of Engineers)
- Awarded travel grant to present research at the IEEE AIoT Conference (2024, Melbourne)
  Otto Mønsteds Fond
- Publication feature: EdgeMark highlighted as a key tool supporting VELUX and Grundfos — Article
  Danish Digital Research Centre (DIREC)
- Ranked 3rd (top 0.02%) nationwide in the entrance examination for M.Sc. programs in electronics
  Iran's national organization of educational testing. Among ∼15,000 participants
- Achieved 2nd place in the Pooyesh Nasir competition
  Department of Electrical Engineering, K. N. Toosi University of Technology
- Nominated for Best Bachelor Project
  Department of Electrical Engineering, K. N. Toosi University of Technology

Personal Highlights

- Fast learner 🚀
- Motivated 🔥
- Creative 🌈
- Diligent 💪
- Optimistic 🌟
- Peaceful 🌸
- Kind, honest 💖
- Nature lover 🌿